Voltage-Controlled Variable Inductor for Fixed-Frequency Critical Conduction Mode Operation

Sihoon Choi, *Member, IEEE*, Jong-Won Shin, *Member, IEEE*, Jun Imaoka, *Member, IEEE*, and Masayoshi Yamamoto, *Member, IEEE*

***Abstract*—This article proposes a novel voltage-controlled variable inductor (VI) for fixing the switching frequency of dc–dc converters in the critical conduction mode (CRM). It overcomes the limitations of conventional current-controlled VIs such as complex winding structures, difficulties in control, and high sensitivity to temperature. In the proposed voltage-controlled VI, the primary side inductance of a coupled inductor is regulated by an auxiliary voltage source on the secondary side. The auxiliary voltage source is implemented by an auxiliary dc–dc converter, which delivers power to the load in parallel with the main converter. The operating principle of the converter system for employing the proposed VI is presented, and a control loop is designed to fix the switching frequency, considering the small-signal model of the auxiliary converter. The validity of the proposed VI is experimentally verified using a prototype CRM boost converter with an output voltage of 400 V and a maximum power of 400 W. The switching frequency is fixed at 100 kHz and the soft switching of the main converter is maintained regardless of the input voltage and output power.**

***Index Terms*—Critical conduction mode (CRM), fixed switching frequency, soft switching, voltage-controlled variable inductor.**

I. Introduction

CRITICAL conduction mode (CRM) is widely applied to dc–dc converters in low-to-medium power applications owing to its soft-switching characteristics and simple control loop design [1], [2]. However, the switching frequency of the CRM converter varies depending on the average inductor current. This variable switching frequency complicates the design of common-mode and differential-mode filters because electromagnetic interference spectrum is distributed over a wide frequency range [3].

Various techniques have been proposed to minimize the variation of the switching frequency. Variable ON-time control for a CRM power factor correction (PFC) boost converter has realized a fixed switching frequency, with a decreased power factor as a tradeoff [4]. The valley skipping method has been applied to limit the range of the switching frequency while securing soft-switching characteristics [2], [5], [6]. However, the switching frequency still varies, quantized by the resonant frequency of the inductor and parasitic capacitances. The hybrid conduction mode control of a three-level boost converter has also been used to fix the switching frequency, reducing the peak inductor current with a quadrangular current waveform [6], [7]. The current waveform becomes triangular at a low input voltage, and the valley skipping technique has been applied to maintain a constant switching frequency.

Variable inductors (VIs) make inductance adjustable and fix the switching frequency. One of the most widely used structures of conventional VIs is the current-controlled configuration, shown in Fig. 1(a) and [8]. The main winding (blue) carries inductor current i_L, which generates a magnetic flux ϕ_L to store energy in the magnetic core during the operation of the converter. The bias current $i_{\rm bias}$ in bias windings (red) generates the flux $\phi_{\rm bias}$ and controls the saturation level of the outer core.

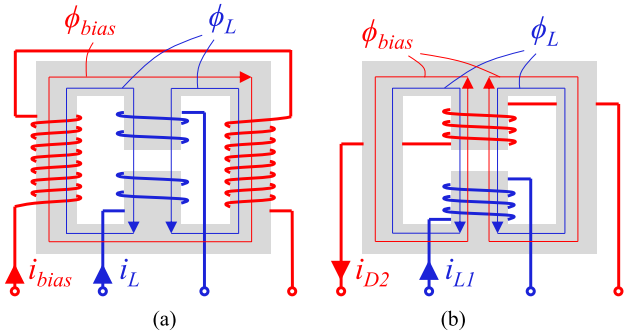


Fig. 1. (a) Conventional current-controlled VI using double E-core. Bias windings (red) saturate the outer core without being coupled to the main winding (blue). (b) Voltage-controlled VI utilized in this article.

Manuscript received 5 April 2022; revised 5 July 2022 and 21 July 2022; accepted 29 July 2022. Date of publication 17 August 2022; date of current version 23 January 2023. This work was supported by the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT under Grant 2019M3F2A1073313, Grant 2021M1A2A2060313, and Grant 2022R1C1C101002711. *(Corresponding author: Jong-Won Shin.)*

Sihoon Choi is with the Graduate School of Engineering, Nagoya University, Nagoya 464-8601, Japan (e-mail: sihoonchoi00@gmail.com).

Jun Imaoka and Masayoshi Yamamoto are with the Institute of Materials and Systems for Sustainability, Nagoya University, Nagoya 464-8601, Japan (e-mail: imaoka@nuee.nagoya-u.ac.jp; m.yamamoto@imass.nagoya-u.ac.jp).

Jong-Won Shin is with the School of Energy Systems Engineering, Chung-Ang University, Seoul 06974, South Korea (e-mail: jwshin@cau.ac.kr).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TIE.2022.3198235.

Digital Object Identifier 10.1109/TIE.2022.3198235

0278-0046 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

The bias windings are wound on two different outer legs, such that ϕ_{bias} is cancelled in the center leg. There is no coupling between the main and bias windings which makes i_{bias} and i_L are independent.

Several converter topologies adopted the conventional VI in Fig. 1(a). An *LLC* converter realized both a constant switching frequency and duty cycle [9], [10]. A single-stage ac–dc converter was developed, featuring a constant dc-link voltage with simpler control [11]. A PFC rectifier in the discontinuous conduction mode (DCM) improved the dynamic characteristics by utilizing conventional VI [12]. A PFC rectifier in CRM had a fixed or quasi-fixed switching frequency [13], [14] and improved the power factor [15]. Performance factors such as transient response [16] and efficiency [17], [18] were also improved by VI.

Other coupled inductor structures, which are different from Fig. 1(a) are also found in the literature. An extra E-core was added to the E-core pair [19], [20]. The bias winding was not required to be on different legs of the core, and this enabled simpler manufacturing. However, the main and auxiliary windings were still coupled and the bias current affected the operation of the main converter. A double toroidal core [21], [22] and a quadruple U-core [23] realized the zero-coupling between the main and bias windings. However, the winding structure was not simple because both the main and bias windings were wound on each core with reverse polarity.

In addition to the low manufacturability due to the complicated winding structure, conventional VIs have further limitation – an additional circuit is required to generate and control the bias current or i_{bias} in Fig. 1(a). A bipolar junction transistor (BJT) with a series resistor [9], [10], [12], [14], [15] is the simplest configuration for i_{bias}, but it consumes power and degrades efficiency. An additional dc–dc converter can also generate i_{bias}, such as a subcircuit of an *LLC* converter [17], a buck converter [22], or a forward converter with a reset winding [24]. Although these converters may experience lower loss than the BJT, they do not contribute to the power delivery. This is because the energy stored in the bias winding is not transferred to the main winding owing to the zero-coupling of conventional VI. Saturating the core causes other issues such as difficulties in control [12] and increased sensitivity to temperature [25].

This study proposes a novel scheme to fix the switching frequency of CRM converter by a voltage-controlled VI with an auxiliary converter. The proposed voltage-controlled VI has two advantages over current-controlled counterpart: 1) it utilizes the typical winding structure of a coupled inductor shown in Fig. 1(b). Because it does not need to realize zero-coupling, the bias winding (red) can be wound on the same leg as the main winding (blue). This configuration is simpler than the conventional one in Fig. 1(a) where the bias winding should be on outer legs; 2) it also guarantees higher manufacturability than the conventional VI. The other advantage of the proposed VI is that it does not saturate the core. This also improves the tolerance of the proposed VI in manufacturing.

The auxiliary converter controls the voltage across the secondary winding of the proposed VI instead of controlling the bias current for saturation, to fix the switching frequency. The auxiliary converter also delivers power to the load in parallel with the main converter. It reduces the power stress of the main converter, current stress of the output capacitor, and ripple of the output voltage. Table I compares the proposed scheme with the conventional ones. One may compare the simplicity of the systems considering modulation methods, auxiliary circuit and windings structure, and magnetic core.

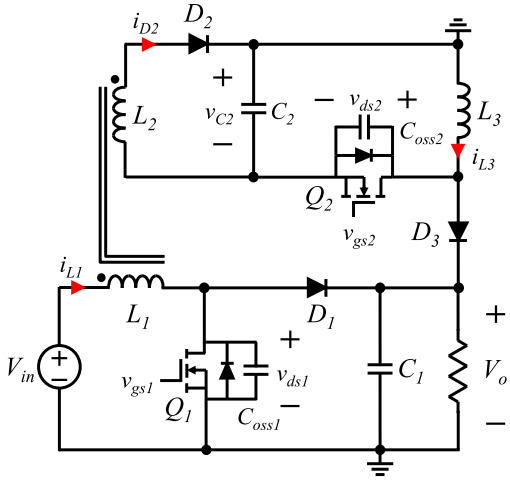


Fig. 2. Proposed voltage-controlled VI with main boost converter (L_1, Q_1, D_1, and C_1) and auxiliary converter (L_2, D_2, C_2, Q_2, L_3, and D_3).

The rest of this article is organized as follows. Section II describes the operation of the converter system using the proposed VI. The operating principle, switching frequency, power distribution between the main and auxiliary converters, and reduction of capacitor current and output voltage ripple are explained. In Section III, small-signal transfer functions are derived and a digital control loop is designed to fix the switching frequency. Section IV presents the experimental verifications. Finally, Section V concludes the article.

II. Converter System With the Proposed Variable Inductor

Fig. 2 shows the main boost converter (L_1, Q_1, D_1, and C_1) and the auxiliary converter (L_2, D_2, C_2, Q_2, L_3, and D_3). Inductors L_1 and L_2 are coupled together as in Fig. 1(b) to build the proposed voltage-controlled VI. Voltages V_{in}, V_o, and v_{C2} denote the input voltage, output voltage, and voltage across C_2, respectively. The auxiliary converter is configured as a buck-boost converter because v_{C2} is controllable whether it is higher or lower than V_o. The self-inductances of the primary and secondary windings are denoted as L_1 and L_2, respectively. The gate-source voltage, drain-source voltage, and the parasitic output capacitance of the switch Q_i are v_{gsi}, v_{dsi}, and C_{ossi}, respectively, where $i = 1$ or 2. The currents i_{L1}, i_{L3}, i_{D2}, and i_{C1} flow through L_1, L_3, D_2, and C_1, respectively.

A. Principle of Operation

Fig. 3 shows the key waveforms of the converter system in Fig. 2. For simplicity, L_1 and L_2 have the same number of turns

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON BETWEEN CONVENTIONAL AND PROPOSED VIs

Reference	[10], [13]	[17]	[22]	[19]	**This work**
Modulation method	Current of the auxiliary winding				**Voltage of the auxiliary capacitor**
Auxiliary circuit	BJT in linear region	Resistor	Buck converter	Customized linear circuit	**Buck-boost converter**
Auxiliary windings	On the two outer legs with the same number of turns [Fig. 1(a)]			On the additional core	**On the center leg [Fig. 1(b)]**
Utilization of magnetic saturation	Yes				**No**
Magnetic core	Double E		Double T	Triple E	**Double E**
Purpose	Fixing f_{sw}	Improving efficiency	Reducing current ripple	Single-stage ac–dc conversion	**Fixing f_{sw}**
Reduction of current stress of output capacitor	No		Yes	No	**Yes**
Reduction of output voltage ripple	No		Yes	No	**Yes**

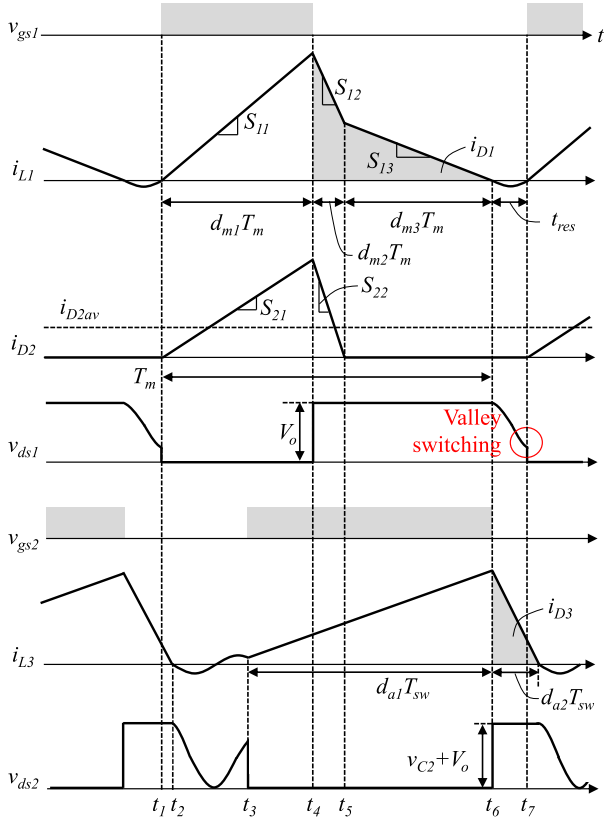


Fig. 3. Key waveforms of the converter system with the proposed VI.

and their inductances are identical to L_s, i.e., $L_1 = L_2 = L_s$. A typical operation of the system comprises the following six intervals, as shown in Fig. 4.

1) Interval 1 [$t_1 < t < t_2$, Fig. 4(a)]: Currents i_{L1} and i_{D2} increase linearly after Q_1 turns ON at t_1 with slopes S_{11} and S_{21} are described as follows:

$$S_{11} = \frac{V_{in} - \alpha v_{C2}}{(1-\alpha^2)L_s} \tag{1}$$

$$S_{21} = \frac{\alpha V_{in} - v_{C2}}{(1-\alpha^2)L_s} \tag{2}$$

where α is the coupling coefficient of the proposed VI.

2) Interval 2 [$t_2 < t < t_3$, Fig. 4(b)]: Current i_{L3} becomes zero at t_2, and C_{oss2} and L_3 begins to resonate. The length of this interval decreases as P_{aux}, the power processed by the auxiliary converter, increases. It is similar to typical dc–dc converters in DCM. In this article, the minimum length of the interval is a quarter of the resonant period of C_{oss2} and L_3 which occurs when P_{aux} is the maximum.
3) Interval 3 [$t_3 < t < t_4$, Fig. 4(c)]: Current i_{L3} increases when Q_2 turns ON at t_3. Time t_3 is determined by the turn-ON time of Q_2 because the turn-OFF instant t_6 is fixed when i_{L1} becomes zero. A detailed explanation of the control of Q_2 is provided in Section III.
4) Interval 4 [$t_4 < t < t_5$, Fig. 4(d)]: When Q_1 turns OFF at t_4, i_{L1} and i_{D2} decrease with slopes S_{12} and S_{22}, respectively, as shown in (3) and (4).

$$S_{12} = \frac{V_{in} - V_o - \alpha v_{C2}}{(1-\alpha^2)L_s} \tag{3}$$

$$S_{22} = \frac{\alpha\,(V_{in} - V_o) - v_{C2}}{(1-\alpha^2)L_s} \tag{4}$$

5) Interval 5 [$t_5 < t < t_6$, Fig. 4(e)]: Interval 5 begins when i_{D2} becomes zero at t_5. In this state, the inductance that i_{L1} experiences becomes the self-inductance of the VI, L_s, because the secondary side of the VI is effectively open. Slope S_{13} is given by (5) which is the same as that of a typical boost converter in the OFF state.

$$S_{13} = \frac{V_{in} - V_o}{L_s} \tag{5}$$

6) Interval 6 [$t_6 < t < t_7$, Fig. 4(f)]: Interval 6 begins when i_{L1} becomes zero at t_6, and C_{oss1} and L_1 resonates. Switch Q_2 simultaneously turns OFF at t_6 to minimize i_{C1}. Though v_{ds2} is as high as $v_{C2} + V_o$, it can be adjusted by properly designing the turns ratio of the coupled inductor or selecting other topologies of the auxiliary converter. In this article, a 1:1 turns ratio of L_1 and L_2 and a buck-boost converter were utilized for the simple design and wide controllable range of v_{C2}.

The main and auxiliary converters operate at the same fixed frequency, and the phase difference between v_{gs1} and v_{gs2} could

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

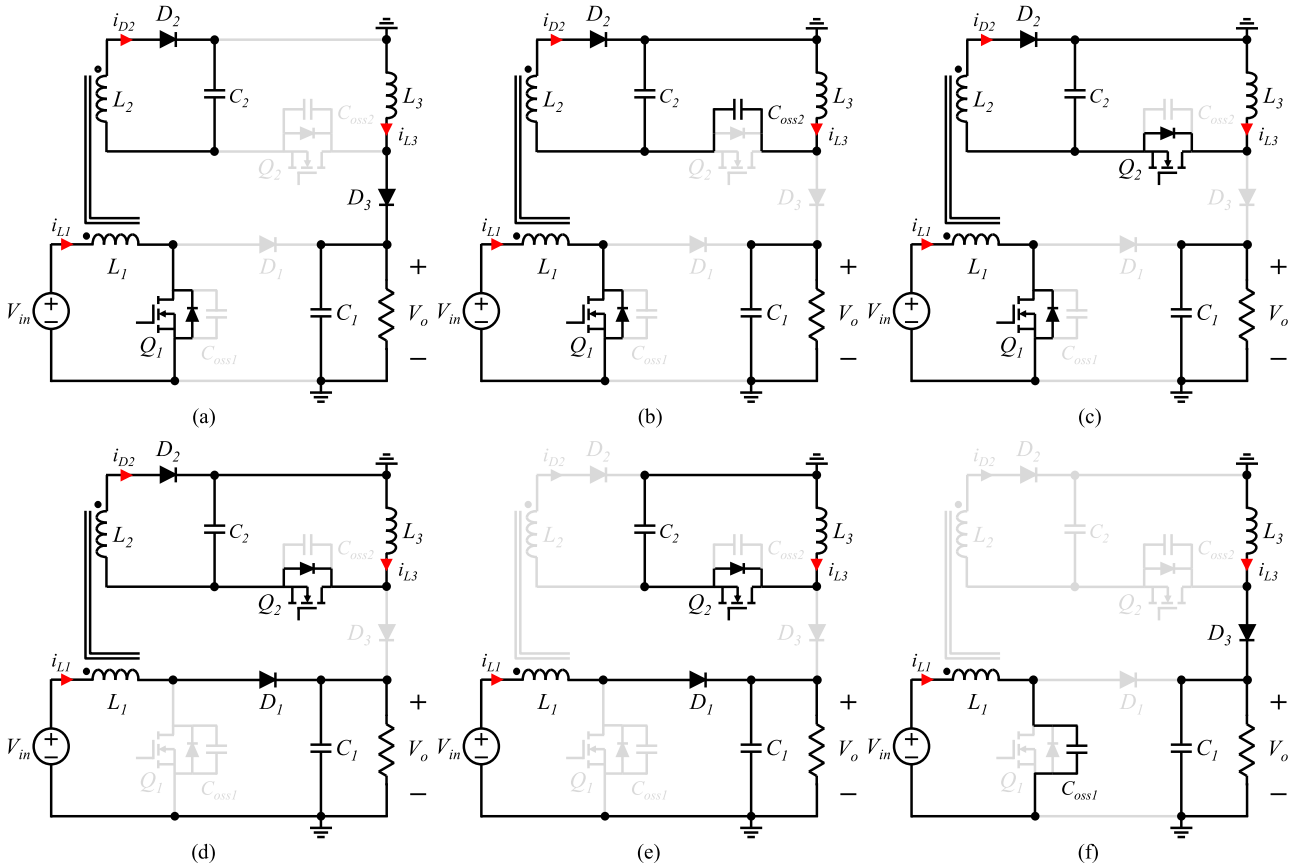


Fig. 4. Operation of the converter system with the proposed VI. (a) Interval 1. (b) Interval 2. (c) Interval 3. (d) Interval 4. (e) Interval 5. (f) Interval 6.

be set for various purposes. In this study, the turn-OFF instant of Q_2, t_6, was synchronized to when i_{L1} became zero to avoid the overlap of i_{D1} (shaded area of i_{L1} in Fig. 3) and i_{D3} (shaded area of i_{L3} in Fig. 3) and minimize the root mean square (rms) i_{C1}.

B. Switching Frequency

The duty ratios related to the main converter, or d_{m1}, d_{m2}, and d_{m3} in Fig. 3, are defined in (6)–(8)

$$d_{m1} = \frac{t_4 - t_1}{T_m} \tag{6}$$

$$d_{m2} = \frac{t_5 - t_4}{T_m} \tag{7}$$

$$d_{m3} = \frac{t_6 - t_5}{T_m} \tag{8}$$

where T_m is defined in (9).

$$T_m = t_6 - t_1 \tag{9}$$

The duty ratios satisfy (10).

$$d_{m1} + d_{m2} + d_{m3} = 1 \tag{10}$$

From the geometry of the i_{L1} and i_{D2} waveforms in Fig. 3, (11) and (12) are obtained.

$$S_{11}d_{m1} + S_{12}d_{m2} + \frac{V_{in} - V_o}{L_s}d_{m3} = 0 \tag{11}$$

$$S_{21}d_{m1} + S_{22}d_{m2} = 0 \tag{12}$$

Solving (10)–(12) yields d_{m1} and d_{m2} as functions of V_{in}, V_o, v_{C2}, and α as in (13) and (14).

$$d_{m1} = \frac{V_o - V_{in}}{V_o} \tag{13}$$

$$d_{m2} = \frac{(V_o - V_{in})(\alpha V_{in} - v_{C2})}{V_o(v_{C2} + \alpha V_o - \alpha V_{in})} \tag{14}$$

The time interval T_m is derived considering the average i_{L1} over a switching period, the output power P_o, and the total efficiency of the main and auxiliary converters η as shown in (15)

$$\frac{1}{2}S_{11}d_{m1}^2 + \frac{1}{2}(2S_{11}d_{m1} + S_{12}d_{m2})d_{m2} + \frac{1}{2}\frac{V_o - V_{in}}{L_s}(1 - d_{m1} - d_{m2})^2 = \frac{P_o}{\eta V_{in} T_m}. \tag{15}$$

The switching period T_{sw} and switching frequency f_{sw} are expressed as a function of v_{C2} in (16), where t_{res} is half the

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

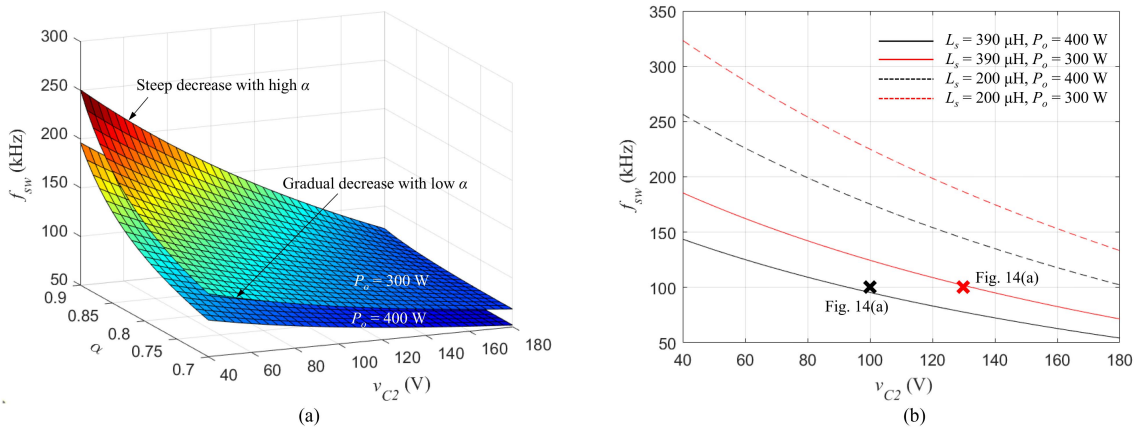


Fig. 5. (a) f_{sw} as a function of α and v_{C2} when $P_o = 400$ W (upper surface) and $P_o = 300$ W (lower surface). (b) f_{sw} when α is 0.85.

resonant period between C_{oss1} and L_s, as in (17).

$$T_{sw} = \frac{1}{f_{sw}} = T_m + t_{res} = \frac{2(\alpha^2-1)L_s P_o(-\alpha V_{in} + \alpha V_o + v_{C2})}{\eta V_{in}(V_o - V_{in})(\alpha V_{in}^2 - \alpha V_{in} V_o - V_{in} v_{C2} + \alpha^2 V_o v_{C2})} + t_{res} \quad (16)$$

$$t_{res} = t_7 - t_6 = \pi\sqrt{L_s C_{oss1}} \quad (17)$$

Assuming that η, V_{in}, V_o, L_s, and C_{oss1} are constants, (16) indicates that f_{sw} depends on α, v_{C2}, and P_o. Fig. 5(a) shows the relationship between f_{sw}, α, and v_{C2} when V_{in}, V_o, L_s, C_{oss1}, and η are 200 V, 400 V, 390 μH, 118 pF, and 95%, respectively. The surface of 300-W P_o is always higher than that of 400-W P_o because f_{sw} is inversely proportional to P_o, assuming that t_{res} is smaller enough than T_m and negligible in (16). As α increases, $\frac{df_{sw}}{dv_{C2}}$ also increases, i.e., the surface of P_o becomes steeper. This implies that a high α reduces the range of v_{C2} to fix f_{sw}. Assuming that L_{eq} is the inductance of the conventional CRM boost converter, the producible L_{eq} ranges from 234.5 μH to 312.7 μH when 300 W $< P_o <$ 400 W, $\alpha = 0.85$, and $t_{res} = 620$ ns. If α is increased to 0.9 with the same conditions, the range of L_{eq} increases – from 208.4 to 375.2 μH. However, too high α may cause excessive f_{sw} and P_{aux}. A detailed description of the power distribution between the main and auxiliary converter is provided in Section II-C.

Equation (16) also shows that f_{sw} is inversely proportional to L_s if t_{res} is negligible. Fig. 5(b) shows various f_{sw} values when α is fixed at 0.85. The solid and dashed lines indicate when L_s is 390 and 200 μH, respectively. The black and red lines indicate $P_o = 400$ W and $P_o = 300$ W, respectively. With L_s fixed, f_{sw} increases as P_o decreases, which is analogous to typical CRM dc–dc converters. The output power P_o changes by the proper design of α and v_{C2} with fixed L_s and f_{sw} from (16). Fig. 6(a) and (b) show P_o versus α and v_{C2} when $v_{C2} = 100$ V and $\alpha = 0.85$, respectively. Fig. 6(a) tells that selecting α affects P_o, i.e., P_o increases when α increases. It is also noted from Fig. 6(b) that f_{sw} is fixed by varying v_{C2} even though P_o changes. The voltage v_{C2} increases when P_o decreases, and f_{sw} is fixed. As the goal of the proposed VI is to fix f_{sw}, it can be implied that v_{C2} varies according to P_o. The experimental results in Section IV or Fig. 14(a) demonstrate the variation in v_{C2}.

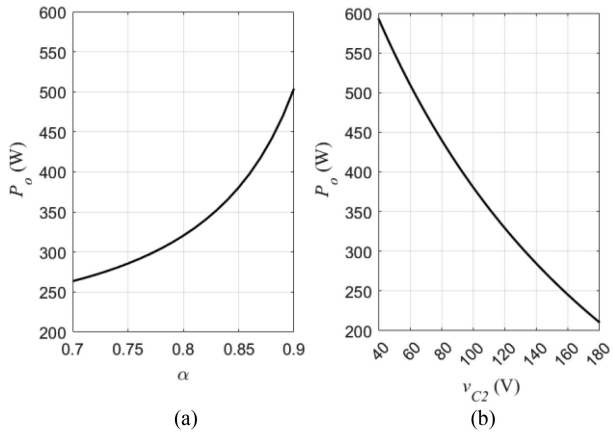


Fig. 6. (a) P_o versus α when $v_{C2} = 100$ V. (b) P_o versus v_{C2} when $\alpha = 0.85$.

C. Power Through the Auxiliary Converter

The power P_{aux} defined in (18) is delivered to the load through the secondary winding of the proposed VI and auxiliary converter as explained in Section II-A

$$P_{aux} = i_{D2av} v_{C2}\,. \quad (18)$$

The current i_{D2av} in (18) is the average i_{D2} over a switching period, as shown in Fig. 3, which satisfies (19)

$$i_{D2av} = \tfrac{1}{2} S_{21} d_{m1}\,(d_{m1} + d_{m2})\,T_{sw}. \quad (19)$$

Fig. 7(a) plots P_{aux} versus v_{C2}, α, and P_o observing (16)–(19), when V_{in}, V_o, L_s, C_{oss1}, and η were 200 V, 400 V, 390 μH, 118 pF, and 95%, respectively. The surface with grey mesh corresponds to $P_o = 400$ W, and the one with the black mesh corresponds to $P_o = 300$ W. They reveal that P_{aux} increases as P_o or α increases. The red and black curves in Fig. 7(b) show P_{aux} when P_o is 300 and 400 W, respectively, while α is 0.85.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

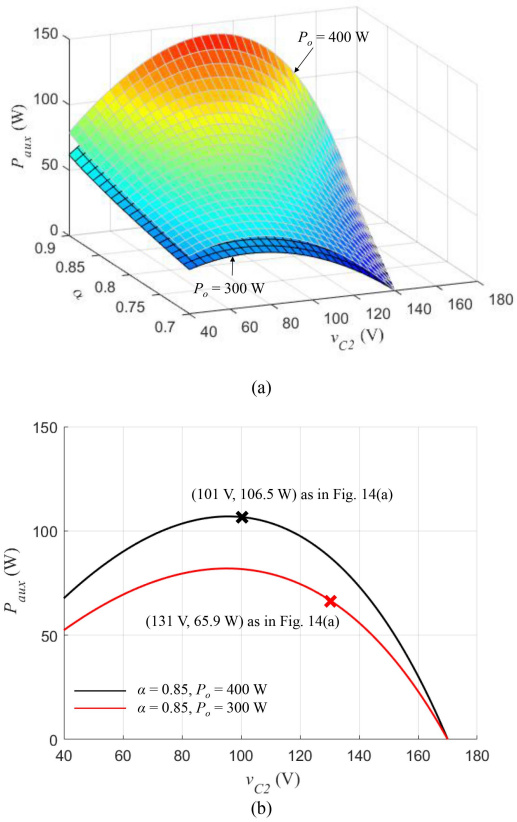


Fig. 7. (a) P_{aux} as a function of α and v_{C2} when $P_o = 400$ W (upper surface with grey mesh) and $P_o = 300$ W (lower surface with black mesh). (b) P_{aux} when α is 0.85.

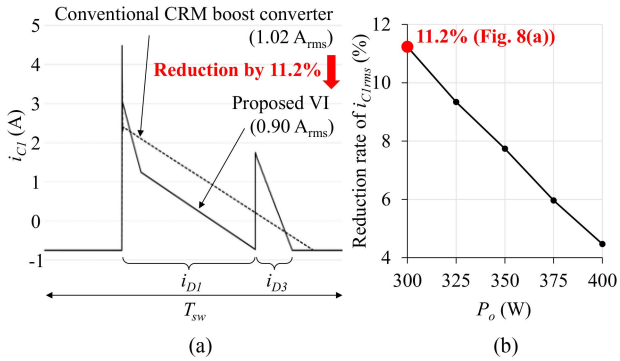


Fig. 8. (a) Simulated waveform of i_{C1} in a switching period with and without the proposed VI when $P_o = 300$ W. (b) Reduction of i_{C1} versus P_o.

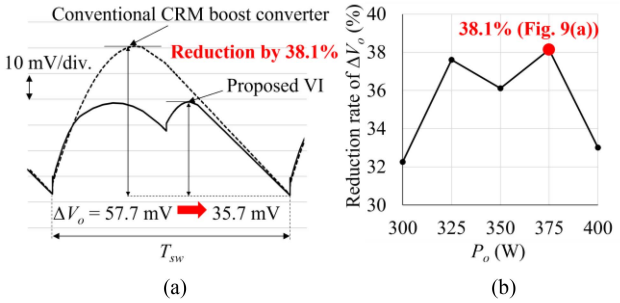


Fig. 9. (a) Simulated waveform of V_o in a switching period with and without the proposed VI when $P_o = 375$ W. (b) Reduction of V_o ripple versus P_o.

The power P_{aux} is proportional to P_o when α and v_{C2} are fixed and t_{res} is negligible.

D. Reduction of Output Capacitor Current and Output Voltage Ripple

In addition to fixing f_{sw}, the auxiliary converter delivers P_{aux} to the load in parallel with the main converter as explained in Section II-C. The current i_{C1} is reduced by delaying v_{gs2} by time T_{d2} with respect to v_{gs1} and avoiding the overlap of i_{D1} and i_{D3} as shown in Fig. 3 and mentioned in Section II-A. An explanation on T_{d2} is provided in Section III. Fig. 8(a) compares the simulated waveforms of i_{C1} (solid line) and the output capacitor current of the conventional CRM boost converter (dashed line) when $P_o = 300$ W. The rms i_{C1} or i_{C1rms} is reduced by 11.2%. The reduction rate of i_{C1rms} when 300 W $< P_o <$ 400 W is in Fig. 8(b).

The output voltage ripple, ΔV_o, is also decreased thanks to the operation of the auxiliary converter. Simulated waveforms of V_o in Fig. 9(a) prove that ΔV_o of the proposed system (solid line) is reduced by 38.1% compared with that of the conventional CRM boost converter (dashed line) when $P_o = 375$ W. The reduction rate of ΔV_o when 300 W $< P_o <$ 400 W is in Fig. 9(b).

III. Control Loop to Fix the Switching Frequency

Fig. 10(a) shows a control concept to fix f_{sw}. Quantities v_{ref}, i_{comp}, and T_{ref} are the digital preset values of the reference output voltage, reference current to detect the zero current of i_{L1}, and reference switching period, respectively. All function blocks in Fig. 10(a) were implemented using a 16-bit digital signal controller, dsPIC33CK32MP102. The analog-to-digital converter (ADC), digital-to-analog converter (DAC), analog comparator, time capture, and pulsewidth modulation (PWM) modules 1 and 2 are the function blocks of dsPIC33CK32MP102. Logic inverters, an AND gate, a falling edge detector, and an set-reset (SR) latch were realized by software. The PI_1 block is a proportional-integral compensation algorithm in the interrupt service routine for controlling v_o. This block generates T_{on1} which is the ON-time value of v_{gs1}. The PI_2 block is another proportional-integral compensation for controlling T_{sw}. Its output is T_{on2} or the ON-time value of v_{gs2}. PWM module 1 generates v_{gs1} which is delayed by time T_{d1} with respect to the signal v_3. Similarly, PWM module 2 outputs v_{gs2} which is delayed by time T_{d2} with respect to v_3.

The proposed control concept does not require any additional circuit components or ADCs, and hardly increases the complexity of the control circuitry. Quantities i_{L1} and v_o are sensed to control Q_1 like a conventional CRM converter as shown in the top shaded area in Fig. 10(a). The proposed control concept utilizes an additional quantity, T_{sw}, However, it is not sensed but derived from i_{L1} which is already sensed to enable the soft switching of Q_1. A time capture block, PI_2 block, and PWM module 2 are the only additional resources to generate v_{gs2} and fix T_{sw} as shown in the bottom shaded area in Fig. 10(a).

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

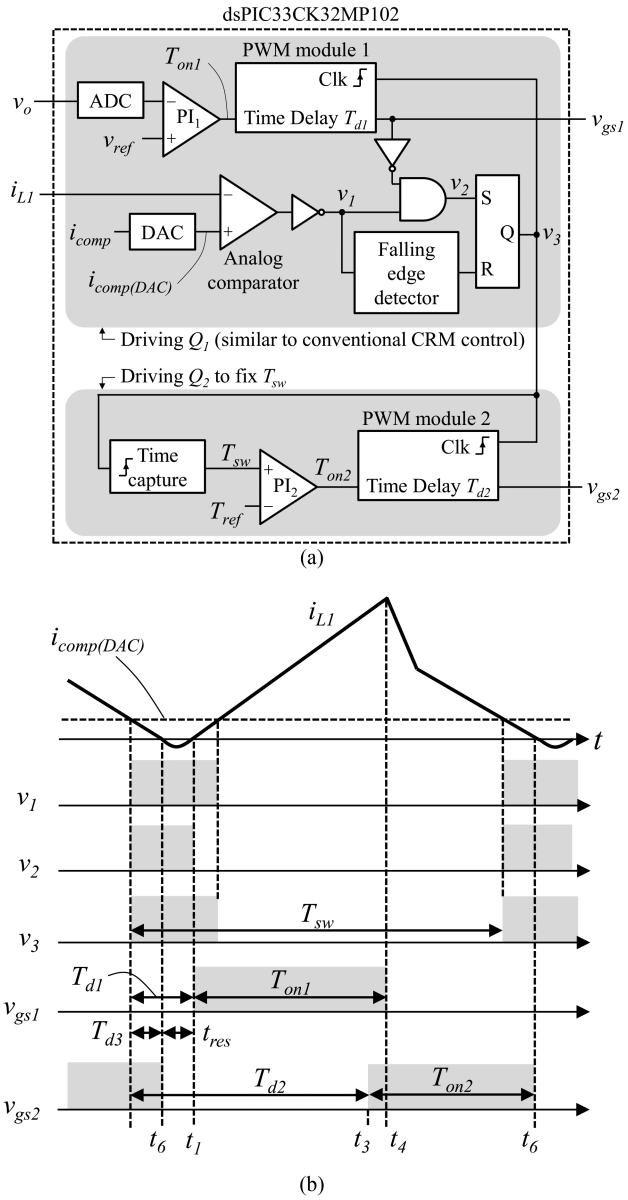


Fig. 10. (a) Control concept to fix f_{sw} in CRM. (b) Timing diagram of the signals in (a).

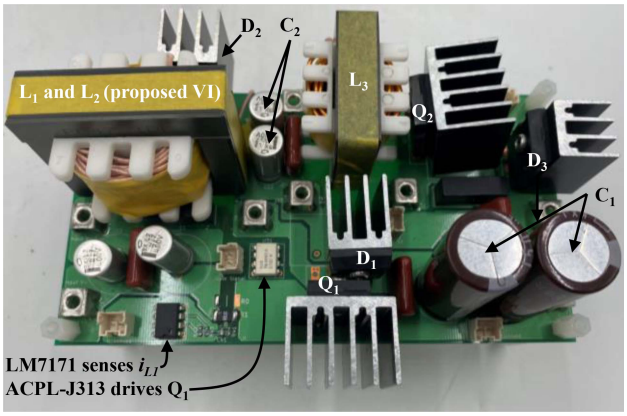


Fig. 11. Prototype hardware used in the experiment.

Fig. 10(b) shows a timing diagram of the proposed control concept. Current i_{L1} is compared with the output of the DAC $i_{comp(DAC)}$ using the analog comparator in Fig. 10(a). The inverted comparator output v_1 is ANDed with inverted v_{gs1} to generate v_2. The rising edge of v_2 sets the output of the SR latch v_3. Using v_2 instead of v_1 as the set signal for the SR latch prevents the start of a new switching period while v_{gs1} is high. Voltages v_{gs1} and v_{gs2} are synchronized with respect to the rising edge of v_3. The time delay T_{d1} is the interval between the rising edges of v_3 and v_{gs1}, are defined as follows:

$$T_{d1} = T_{d3} + t_{res} \tag{20}$$

where t_{res} is given by (17). Time T_{d3} in (20) is given by (21).

$$T_{d3} = -\frac{i_{comp(DAC)}}{S_{13}}. \tag{21}$$

Note that T_{d3} is positive because $i_{comp(DAC)}$ is a small positive value to achieve noise immunity, and S_{13} is negative as shown in Fig. 3 and (5). Turning ON Q_1 at t_1 by T_{d1} reduces the switching loss because v_{ds1} has a minimum value at t_1.

Switch Q_2 should be turned OFF at t_6 when i_{L1} becomes zero as shown in Figs. 3 and 10(b), which is to minimize the current stress of C_1 as mentioned in Section II-A. Thus, the falling edge of v_{gs2} is fixed to t_6 by setting T_{d2} as the interval between the rising edges of v_3 and v_{gs2} as in (22)

$$T_{d2} = T_{sw} - T_{on2} + T_{d3}. \tag{22}$$

The time capture block in Fig. 10(a) detects T_{sw} by monitoring the rising edge of v_3. It transfers T_{sw} to the positive terminal of the PI_2 block, whereas T_{ref} is fed to the negative terminal. It is noted that the configuration is different from the typical feedback where the reference is fed to the positive terminal. It is to compensate the negative sign of the small-signal transfer function from d_{a1} (duty ratio of v_{gs2}) to v_{C2} in (23)

$$\frac{\hat{v}_{C2}}{\hat{d}_{a1}} = -\frac{2V_{C2}}{D_{a1}} \frac{1}{1 + s \Big/ \frac{D_{a1}^2 T_{sw}}{2C_2 L_3}}. \tag{23}$$

In (23), s is the Laplace operator, and V_{C2} and D_{a1} are the dc components of v_{C2} and d_{a1}, respectively.

The small-signal transfer function from v_{C2} to T_{sw} is obtained from (16) as in (24).

$$\frac{\hat{T}_{sw}}{\hat{v}_{C2}} = \frac{2\left(1-\alpha^2\right) L_s P_o \alpha^3 V_o^2}{\eta V_{in}\left(\alpha V_{in}^2 - \alpha V_{in} V_o - V_{in} V_{C2} + \alpha^2 V_o V_{C2}\right)^2}. \tag{24}$$

Combining (23) and (24) yields the transfer function from d_{a1} to T_{sw} in (25).

$$\frac{\hat{T}_{sw}}{\hat{d}_{a1}} = \frac{\hat{v}_{C2}}{\hat{d}_{a1}} \cdot \frac{\hat{T}_{sw}}{\hat{v}_{C2}}. \tag{25}$$

IV. Experimental Verification

The converter system with the proposed VI was built on a 15 × 8 cm 2-layer printed circuit board (PCB) as shown in Fig. 11. The circuit elements and parameters are listed in Table II. Though the experimental conditions are for a microconverter in a dc grid, the application of the proposed VI and its control method are not limited to this. The advantages of the proposed concept become apparent in high-voltage or high-frequency applications in which the switching loss and noise are severe.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

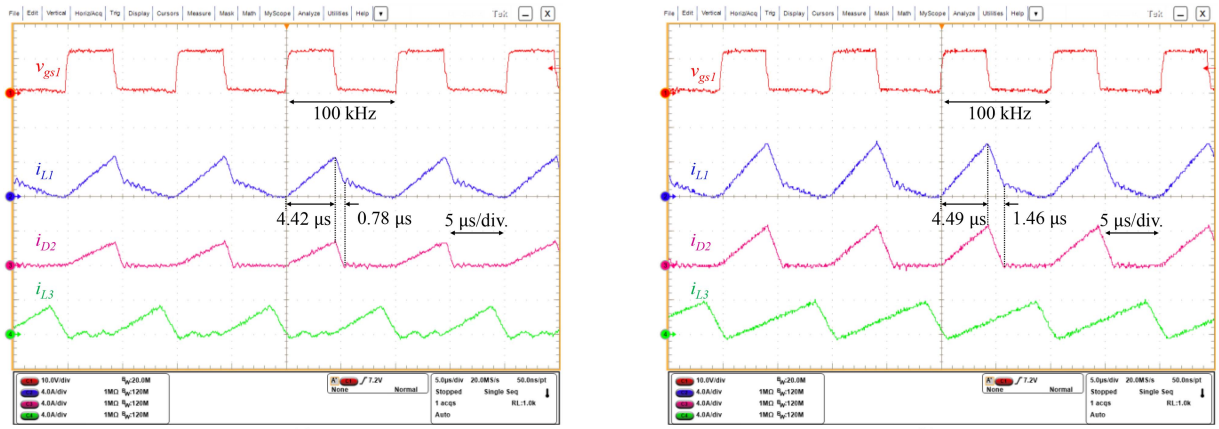


Fig. 12. Experimental current waveforms of the converter system using the proposed VI. Red trace is v_{gs1} with 10 V/div. Blue trace is i_{L1} with 4 A/div. Magenta trace is i_{D2} with 4 A/div. Green trace is i_{L3} with 4 A/div. (a) P_o is 300 W. (b) P_o is 400 W.

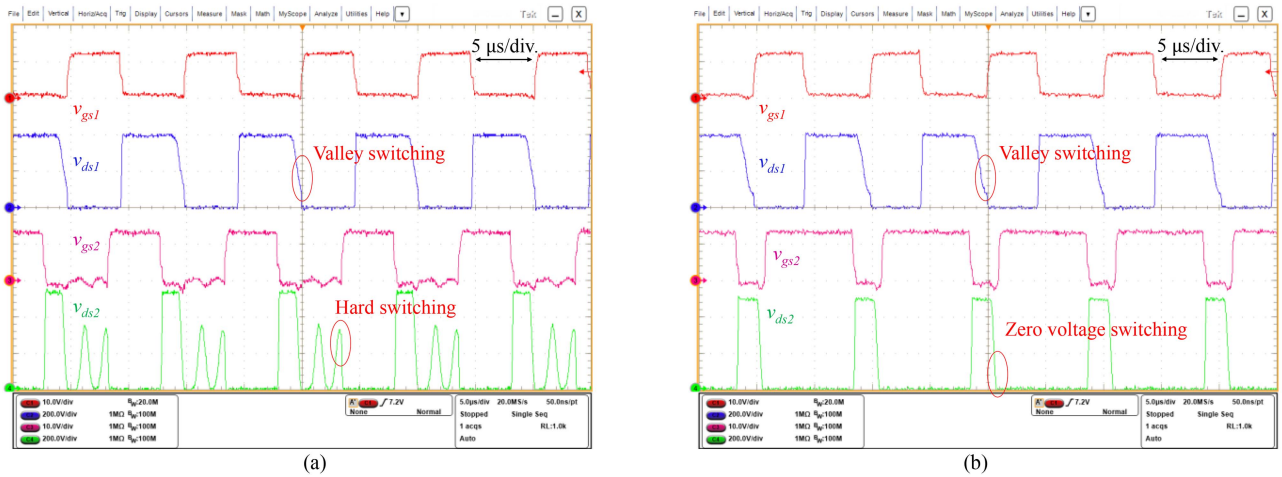


Fig. 13. Experimental voltage waveforms of the converter system using the proposed VI. Red trace is v_{gs1} with 10 V/div. Blue trace is v_{ds1} with 200 V/div. Magenta trace is v_{gs2} with 10 V/div. Green trace is v_{ds2} with 200 V/div. (a) P_o is 300 W. (b) P_o is 400 W.

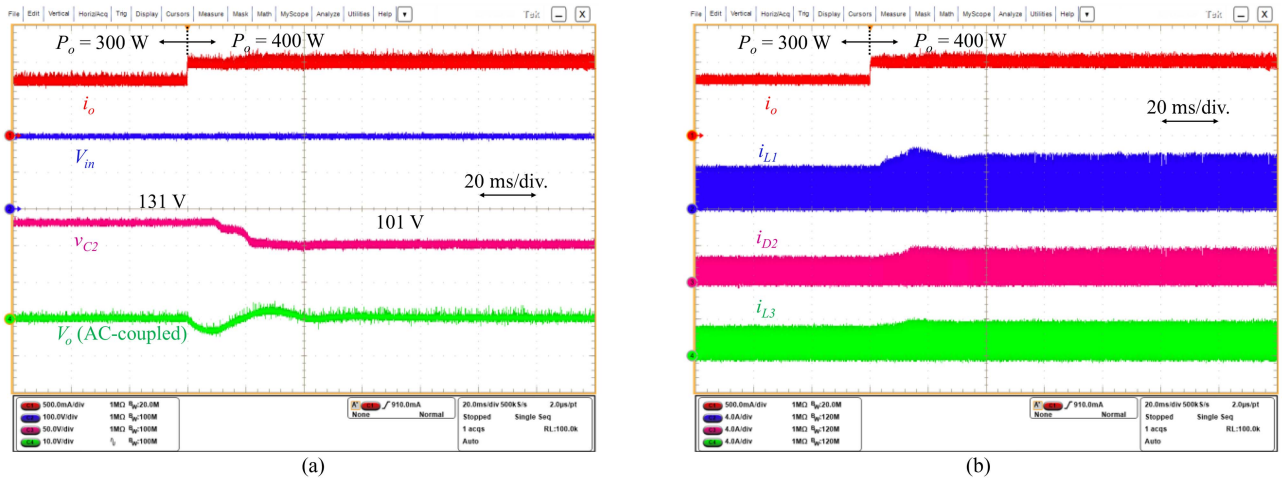


Fig. 14. Transient response of the converter system using the proposed VI when P_o jumps from 300 to 400 W. Red trace is i_o with 500 mA/div. (a) Blue trace is V_{in} with 100 V/div. Magenta trace is v_{C2} with 50 V/div. Green trace is ac-coupled V_o with 10 V/div. (b) Blue trace is i_{L1} with 4 A/div. Magenta trace is i_{D2} with 4 A/div. Green trace is i_{L3} with 4 A/div.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

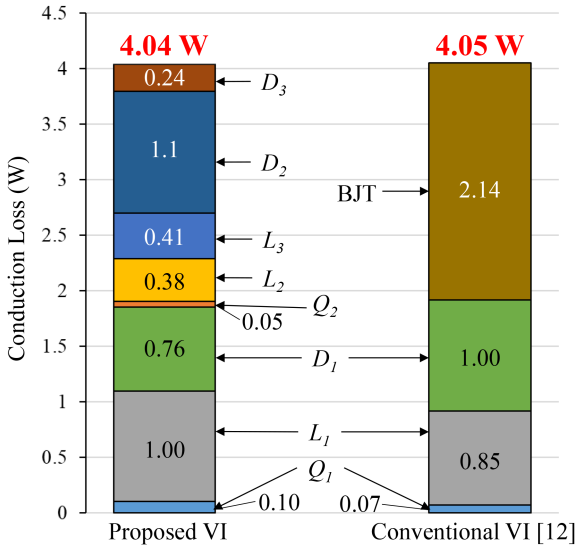


Fig. 15. Estimated conduction loss breakdown of the proposed VI and conventional VI [12] when $P_o = 400$ W.

TABLE II
CONVERTER SPECIFICATION

V_{in}	200 V
V_o	400 V
L_s	390 μH
L_3	205 μH
α	0.85
$P_{o,max}$	400 W
C_1	300 μF
C_2	10 μF
Q_1 and Q_2	SCT3022AL
$C_{oss1\ and}\ C_{oss2}$	118 pF
D_1, D_2, $_{and}$ D_3	C3D20060D
T_{ref}	10 μs (reciprocal of 100 kHz)

The power P_{aux} ranges from approximately 65.9 W [indicated by a red cross in Fig. 7(b)] to 106.5 W [indicated by a black cross in Fig. 7(b)]. Inductor L_3 was designed to operate the auxiliary converter in CRM when $P_o = 400$ W and in DCM when $P_o < 400$ W.

The proposed VI was built using two PC47EI50Z ferrite cores manufactured by TDK. Both the main and auxiliary windings had 28 turns and were wound on the center leg. An air gap of 0.75 mm was added to the center and outer legs. The resultant L_s and α were 390 μH and 0.85, respectively, as in Table II.

Fig. 12 shows the experimental current waveforms of i_{L1} (blue trace), i_{D2} (magenta), and i_{L3} (green) with v_{gs1} (red). The waveshapes were similar to those shown in Fig. 3. The measured $d_{m1}T_m$ and $d_{m2}T_m$ were 4.42 μs and 0.78 μs, respectively, when $P_o = 300$ W as in Fig. 12(a). They were 4.49 μs and 1.46 μs when $P_o = 400$ W as in Fig. 12(b). These match well with those calculated from (9) and (10) which were 4.66 μs and 0.83 μs at 300 W and 4.66 μs and 1.34 μs at 400 W. The waveshape of i_{L3} implies that the auxiliary converter operated in CRM when $P_o = 400$ W and in DCM when $P_o = 300$ W as previously mentioned.

Fig. 13 displays experimental voltage waveforms of v_{ds1} (blue trace), v_{gs2} (magenta), and v_{ds2} (green). The waveform of v_{gs1} (red) is the same as that in Fig. 12 for clarity. Switch Q_1 was turned ON with valley switching regardless of P_o. In Fig. 13(a), Q_2 experienced a hard turn-ON, because the auxiliary converter operated in DCM when $P_o < 400$ W. However, Q_2 was turned ON with zero voltage when $P_o = 400$ W as in Fig. 13(b).

Fig. 14 presents the transient response of the converter system using the proposed VI when P_o jumped from 300 to 400 W. The red trace indicates the output current i_o through the load resistance. Fig. 14(a) shows the voltage waveforms, such as V_{in} (blue trace), v_{C2} (magenta), and ac-coupled V_o (green). In Fig. 14(b), current waveforms, such as i_{L1} (blue), i_{D2} (magenta), and i_{L3} (green) are shown. The converter system was demonstrated to be stable with the control scheme explained in Section III. The measured magnitude of v_{C2} such as 131 and 101 V are marked in Fig. 5(b) as red and black cross, respectively.

The proposed converter system features comparable power loss to the conventional system using the voltage-controlled VI. Fig. 15 shows the estimated conduction loss breakdowns of the proposed and conventional VI in [12] when $P_o = 400$ W. The losses were calculated under the same conditions defined in Table II for fair comparison. The total losses of both VIs are similar to each other. Losses of L_1 and Q_1 of the proposed VI were larger than those of [12] because of the higher peak inductor current. The sum of the losses of D_1 and D_3 in the proposed scheme was the same with the loss of D_1 in [12] since P_o was the same. The loss of the BJT, 2.14 W, was calculated by multiplying the bias current (0.43 A) and supply voltage (5 V) based on [12], assuming $L_s = 234.5$ μH, $f_{sw} = 100$ kHz, and $t_{res} = 620$ ns. The loss of the proposed system is not larger than that of the conventional counterpart despite the additional circuit components in the auxiliary converter. This is because L_2, D_2, C_2, Q_2, L_3, and D_3 are effectively a parallel converter to the main converter and minimize the impact on efficiency degradation.

V. CONCLUSION

A novel voltage-controlled VI was proposed for the constant-frequency CRM operation of dc–dc converters. The secondary-side voltage of the proposed VI was modulated by an auxiliary converter, and the primary-side inductance was modulated to fix the switching frequency. The proposed VI did not require a complicated core, winding structure, or core saturation. The main converter experienced soft switching by CRM operation with a fixed switching frequency. The auxiliary converter contributed to the power transmission between the source and load in parallel with the main converter, which reduces power stress of the main converter, current stress of the output capacitor, and the output voltage ripple.

The operation principle, duty ratios, power distribution between the main and auxiliary converters, and small-signal transfer functions to establish closed-loop control were presented. The validity of the proposed VI was proved by a prototype converter system with a switching frequency of 100 kHz and a maximum output power of 400 W.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] R. Erickson and D. Maksimović, *Fundamentals of Power Electronics*, 3rd ed., Switzerland: Springer, 2020, pp. 889–892.

[2] L. Huber, B. Irving, and M. M. Jovanović, "Effect of valley switching and switching-frequency limitation on line-current distortions of DCM/CCM boundary boost PFC converter," *IEEE Trans. Power Electron.*, vol. 24, no. 2, pp. 339–347, Feb. 2009.

[3] F. Yang, X. Ruan, Q. Ji, and Z. Ye, "Input differential-mode EMI of CRM boost PFC converter," *IEEE Trans. Power Electron.*, vol. 28, no. 3, pp. 1177–1188, Mar. 2013.

[4] K. Yao, Y. Wang, J. Guo, and K. Chen, "Critical conduction mode boost PFC converter with fixed switching frequency control," *IEEE Trans. Power Electron.*, vol. 33, no. 8, pp. 6845–6857, Aug. 2018.

[5] J. Shin, G. Seo, B. Cho, and K. Lee, "Modeling and implementation of digital control for critical conduction mode power factor correction rectifier," in *Proc. IEEE Workshop Control Model Power Electron.*, 2012, pp. 1–8.

[6] M. Lee and J. Lai, "Fixed-frequency hybrid conduction mode control for three-level boost PFC converter," *IEEE Trans. Power Electron.*, vol. 36, no. 7, pp. 8334–8346, Jul. 2021.

[7] M. Lee, J. Kim, and J. Lai, "Digital-based critical conduction mode control for three-level boost PFC converter," *IEEE Trans. Power Electron.*, vol. 35, no. 7, pp. 7689–7701, Jul. 2020.

[8] D. Medini and S. Ben-Yaakov, "A current-controlled variable inductor for high frequency resonant power circuits," in *Proc. IEEE Appl. Power Electron. Conf.*, vol. 1, 1994, pp. 219–225.

[9] Y. Wei, Q. Luo, X. Du, N. Altin, J. M. Alonso, and A. Mantooth, "Analysis and design of the LLC resonant converter with variable inductor control based on time-domain analysis," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5432–5443, Jul. 2020.

[10] Y. Wei, Q. Luo, and A. Mantooth, "Comprehensive analysis and design of LLC resonant converter with magnetic control," *CPSS Trans. Power Electron. Appl.*, vol. 4, no. 4, pp. 265–275, Dec. 2019.

[11] Y. Wei, Q. Luo, J. M. Alonso, and A. Mantooth, "A magnetically controlled single-stage AC/DC converter," *IEEE Trans. Power Electron.*, vol. 35, no. 9, pp. 8872–8877, Sep. 2020.

[12] Y. Wei, Q. Luo, J. Wang, and S. Pengju, "Analysis and design of the DCM operation boost PFC converter with magnetic control," *IET Power Electron.*, vol. 12, no. 14, pp. 3697–3706, Dec. 2019.

[13] K. Yao, Z. Zhang, J. Yang, J. Liu, J. Li, and F. Shao, "Quasi-fixed switching frequency control of CRM boost PFC converter based on variable inductor in wide input voltage range," *IEEE Trans. Power Electron.*, vol. 36, no. 2, pp. 1814–1827, Feb. 2021.

[14] Z. Zhang et al., "All-fixed switching frequency control of CRM boost PFC converter based on variable inductor in a wide input voltage range," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2019, pp. 1434–1441.

[15] C. Lim, Y. Jeong, and G. Moon, "Phase leading input capacitor compensation using variable inductor with high efficiency in a CRM boost PFC," in *Proc. IEEE 3rd Int. Future Energy Electron. Conf.*, 2017, pp. 852–856.

[16] S. Ahsanuzzaman, T. McRae, M. Peretz, and A. Prodic, "Low-volume buck converter with adaptive inductor core biasing," in *Proc. 27th Annu. IEEE Appl. Power Electron. Conf. Expo.*, 2012, pp. 335–339.

[17] C. Lim, J. Kim, Y. Jeong, D. Kim, H. Youn, and G. Moon, "A high efficiency critical mode boost PFC using a variable inductor," in *Proc. IEEE Int. Power Electron. Motion Control Conf.*, 2016, pp. 2792–2797.

[18] Y. Wei, N. Altin, Q. Luo, and A. Nasiri, "A high efficiency decoupled on-board battery charger with magnetic control," in *Proc. 7th Int. Conf. Renewable Energy Res. Appl.*, 2018, pp. 920–925.

[19] Y. Hu, L. Huber, and M. M. Jovanovic, "Single-stage universal-input AC/DC LED driver with current-controlled variable PFC boost inductor," *IEEE Trans. Power Electron.*, vol. 27, no. 3, pp. 1579–1588, Mar. 2012.

[20] E. Bitencourt, M. Cosetin, I. Vegner, and R. do Prado, "A ferromagnetic based variable inductor analysis and design methodology," in *Proc. IEEE 13th Braz. Power Electron. Conf. 1st Southern Power Electron. Conf.*, 2015, pp. 1–5.

[21] A. Kislovski, "Quasi-linear controllable inductor," in *Proc. IEEE*, vol. 75, no. 2, pp. 267–269, Feb. 1987.

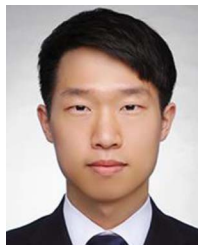
[22] M. W. Beraki, J. P. F. Trovão, M. S. Perdigão, and M. R. Dubois, "Variable inductor based bidirectional DC–DC converter for electric vehicle," *IEEE Trans. Veh. Technol.*, vol. 66, no. 10, pp. 8764–8772, Oct. 2017.

[23] J. Alonso, M. Perdigão, M. Dalla Costa, S. Zhang, and Y. Wang, "Analysis and experimentation of the quad-U variable inductor for power electronics applications," *IET Power Electron.*, vol. 11, no. 14, pp. 2330–2337, Nov. 2018.

[24] M. S. Perdigao, J. M. Alonso, M. A. Dalla Costa, and E. S. Saraiva, "Comparative analysis and experiments of resonant tanks for magnetically controlled electronic ballasts," *IEEE Trans. Ind. Electron.*, vol. 55, no. 9, pp. 3201–3211, Sep. 2008.

[25] J. Imaoka, T. Aoki, K. Shigematsu, and M. Yamamoto, "Temperature dependency modeling simulation and experimental validations of DC superimposition characteristics for gapped-ferrite inductor including saturation region," in *Proc. IEEE 29th Int. Symp. Ind. Electron.*, 2020, pp. 1466–1473.

[26] J. Sun, D. Mitchell, M. Greuel, P. Krein, and R. Bass, "Averaged modeling of PWM converters operating in discontinuous conduction mode," *IEEE Trans. Power Electron.*, vol. 16, no. 4, pp. 482–492, Jul. 2001.



Sihoon Choi (Member, IEEE) received the B.S. degree in energy systems engineering from Chung-Ang University, Seoul, South Korea, in 2019. He is currently working toward the M.S. degree in electrical engineering with Nagoya University, Nagoya, Japan.

From 2019 to 2020, he was an Associate Research Engineer with LG Innotek, Seoul, South Korea. His research interests include power conversion system, modeling of magnetic components, and SiC power module packaging.



Jong-Won Shin (Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Seoul National University, Seoul, South Korea, in 2006 and 2013, respectively.

From 2013 to 2015, he was a Postdoctoral Researcher with Virginia Tech, Blacksburg, VA, USA. From 2015 to 2018, he was a Senior Scientist with the Electronics Research Department, Toyota Research Institute of North America, Ann Arbor, MI, USA. He joined Chung-Ang University, Seoul, South Korea in 2018, where he is currently an Associate Professor. Since 2021, he has also been a Visiting Associate Professor with Nagoya University, Nagoya, Japan. His research interests include power conversion, energy management, and power semiconductor packaging.



Jun Imaoka (Member, IEEE) received the M.S. and Ph.D. degrees in electronic function and system engineering from Shimane University, Matsue, Japan, in 2013 and 2015, respectively.

From October 2015 to March 2018, he was an Assistant Professor with Kyushu University, Fukuoka, Japan. From April 2018 to March 2021, he was an Assistant Professor with Nagoya University, Nagoya, Japan. He is currently an Associate Professor with the Institute of Materials and Systems for Sustainability (IMaSS), Nagoya University. His research interests include the design of integrated magnetic components, modeling for high-power-density power converters, thermal management for power converters, magnetic material application, and EMI of switching power supply.



Masayoshi Yamamoto (Member, IEEE) received the M.S. and Ph.D. degrees in science and engineering from Yamaguchi University, Yamaguchi, Japan, in 2000 and 2004, respectively.

From 2004 to 2005, he was with Sanken Electric Co., Ltd., Saitama, Japan. From 2006 to 2017, he was an Associate Professor with the Interdisciplinary Faculty of Science and Engineering, Shimane University, Matsue, Japan. He is currently a Professor with the Institute of Materials and Systems for Sustainability (IMaSS), Nagoya University, Nagoya, Japan. His research interests include automotive power electronics, power module packaging, EMI of switching power supply, and wireless power transfer.

Authorized licensed use limited to: Sihoon Choi. Downloaded on March 31,2023 at 10:31:01 UTC from IEEE Xplore. Restrictions apply.